DAYANAND SAGAR ACADEMY Attendance ERP

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This document explains every feature in plain language and shows how to demo the system end-to-end. It uses mock data with localStorage persistence, so nothing else is required beyond Node and a browser.

> If you only want to run the app: follow “Setup” → “Run locally”. Everything else is feature explanations and demo guides.

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1) Setup

- Requires **Node 20.19.0+** (pinned in `.nvmrc`). On Windows use nvm-windows; on macOS/Linux use nvm. Commands:

```
cd "c:\Users\smart\Desktop\attendance erp app"
nvm install 20.19.0
nvm use 20.19.0
npm install
npm run dev   # http://localhost:5173 (auto-bumps port if busy)
```

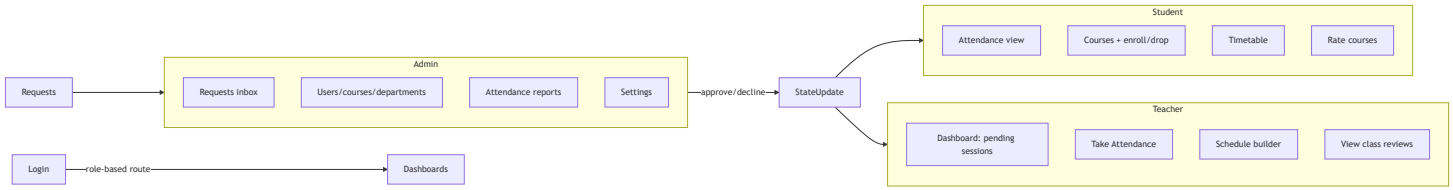
Build/preview:

```
npm run build
npm run preview
```

2) High-level map

- Three roles: **Student**, **Teacher**, **Admin**. Each sees tailored navigation and permissions.
- Data lives in Redux slices and is saved to `localStorage` (keys prefixed with `nhu-`). Refreshing keeps state; clearing storage resets to seed data.
- Mock-only app: all “requests” (approvals) are stored in Redux; no real emails are sent.

Mermaid overview:



3) How to demo (scripted)

Use the built-in demo logins (on the login page):

- Student: `student@nhu.edu` / `student123`
- Teacher: `teacher@nhu.edu` / `teacher123`
- Admin: `admin@nhu.edu` / `admin123`

Suggested walkthrough (10–15 minutes):

1. **Student journey**
   - Log in as student.
   - Dashboard: point out overall attendance %, today’s classes, per-course chips.
   - Attendance page: filter by course and date; show calendar legend (Present/Late/Absent/Pending/No Class).
   - Courses: enroll in an optional course, submit a drop request for an optional one, rate a course, view all reviews.
   - Timetable: switch days on mobile selector, show grid on desktop.
2. **Teacher journey**
   - Log in as teacher.
   - Dashboard: see “Pending Attendance” count; click “Take Attendance”.

- Take Attendance: pick session, mark all present, change one student to late, save. Show past class edit flow (requires admin approval).
- Schedule: add an extra session (overlap protection), browse calendar; open edit/delete on a past session to trigger approval request.
- Classes: export CSV, open course reviews dialog.

3. **Admin journey**
   - Log in as admin.
   - Requests: approve/decline a course drop, student registration, attendance update, or session edit (statuses update in-line).
   - Users: view students/teachers; Structure: departments/batches; Courses: rating column and edit dialog for any review.
   - Reports: open attendance report view; Settings: confirm mock nature and seed data.

4) Features by role

Shared

- **Authentication**: role-based redirect, persisted session (`nhu-auth-user`).
- **Notifications**: in-app list powered by `notificationsSlice` (created when requests are submitted).
- **Requests**: `requestsSlice` stores approval items with statuses `pending | approved | declined` and approvers (admin by default).

Student (`pages/student/*`)

- **Dashboard**: overall attendance %, today's classes (time, room, teacher), absences this month, attendance by course with traffic-light chips.
- **Attendance**: filter by course/date range; table shows date/time/teacher/status/remarks; calendar heatmap with status legend; overall % indicator.
- **Courses**:
  - Enroll optional courses instantly (adds to enrollments).
  - Request drop (creates admin approval request + notification).
  - Rate/feedback per course (submit or edit own review); view all reviews; average rating displayed on cards.
- **Timetable**: weekly view; mobile day selector and desktop 6-column grid; shows date/time/room/teacher per session.

Teacher (`pages/teacher/*`)

- **Dashboard**: counts courses, today's classes, pending attendance (sessions today without records); quick CTA to "Take Attendance".
- **Take Attendance**:
  - Session selector (persisted as active session).
  - Live gate: can only mark during ongoing class; past sessions go through approval flow.
  - Mark-all shortcuts; per-student Present/Late/Absent buttons; save writes records and marks session complete.
  - Past classes list: "Edit Attendance" opens dialog; submission sends "attendance-update" request to admin; shows request status badge.
- **Classes**: for each course see students count, sessions count, average attendance %, preview three sessions, export CSV roster attendance %, view reviews dialog (read-only).
- **Schedule**:
  - Add sessions with frequency (once/twice/weekdays), overlap validation, session type tags, notes.
  - Calendar month grid highlighting weekends; upcoming sessions list.
  - Edit/delete: past sessions require admin approval via "session-edit" request; status chip shown per session.

Admin (`pages/admin/*`)

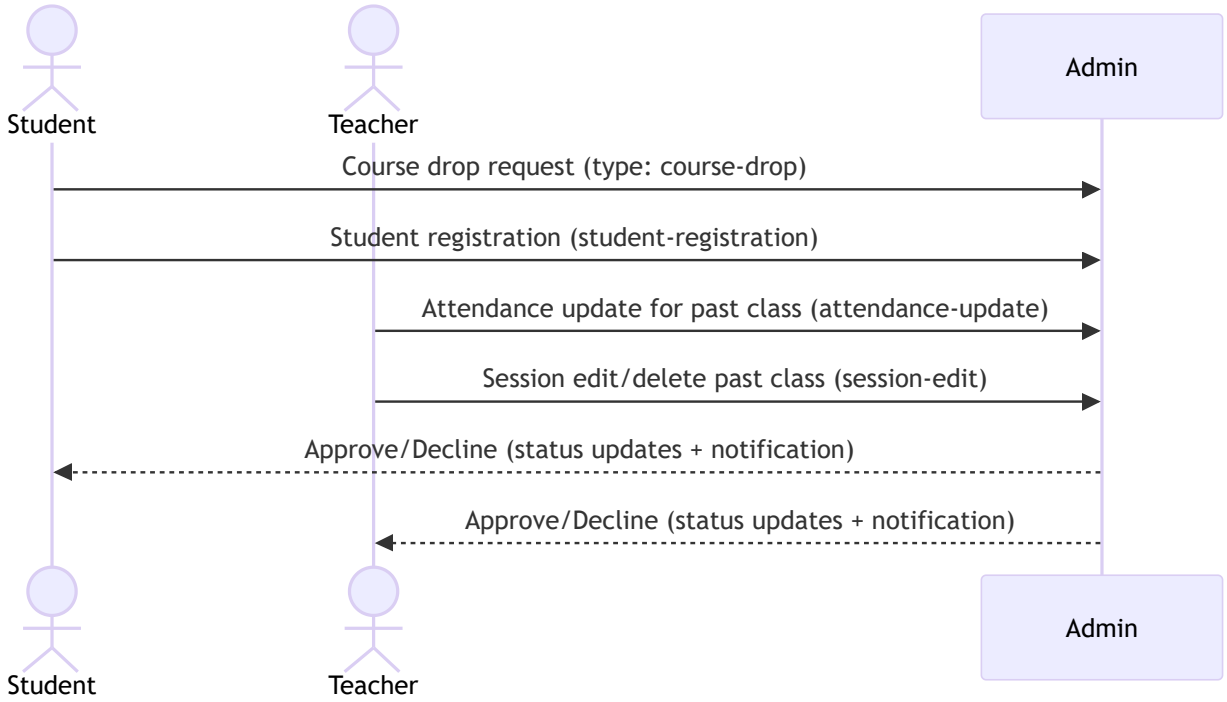
- **Requests**: central inbox for all request types (student/teacher registration, course drop, attendance-update, session-edit). Actions: approve or decline; status reflects immediately.
- **Users**: view/manage student and teacher lists.
- **Structure**: manage departments and batches (used in registration form filters).
- **Courses**: table includes average rating column; admin can view and edit any review in dialog; manage course metadata.
- **Attendance Reports**: aggregated attendance summaries and per-course stats.
- **Settings**: clarify mock mode and seeds (no external services).

Reviews & Ratings (all roles, different permissions)

- Students: submit or edit their own course review (rating + optional comment); see all reviews.
- Teachers: view reviews for their courses (read-only).
- Admin: view and edit any review in course dialog.
- Every course card (student/teacher/admin) shows average rating and count.

5) Approval workflows

All approvals are in-app (Redux). Notifications are created for admins on submission.



Behavior notes:

- **Pending** stays until admin acts. No auto-expiry.
- **Approved** requests imply the admin will perform the change; mock app records status only (no external side effects beyond stored status and optional edits).
- **Declined** keeps the request visible with final status for traceability.

6) Edge cases and guardrails

- **Attendance editing**: Live classes can be marked directly. Past classes force an approval request and show a status chip so teachers know if it was processed.
- **Schedule overlaps**: Creating sessions checks time conflicts; blocked with a warning.
- **Role defaults**: Unknown routes redirect to each role's default dashboard; unauthenticated users go to login.
- **Data persistence**: Clearing `localStorage` resets to seed data. Good for demo resets.
- **Ratings**: Students edit only their own review; admin can edit any; teacher is read-only.
- **Enroll/drop**: Core courses cannot be dropped; optional courses create a drop request instead of immediate removal.

7) Data model (plain language)

- **Users**: student, teacher, admin. Stored in `adminSlice` mock data; auth stores current user and `status` / `error`.
- **Courses**: belong to departments/batches; have teacherId, credits, schedule summary, and optional flag.
- **Sessions**: class instances with courseId, teacherId, date, start/end, room, type, notes.
- **Enrollments**: studentId + courseId (+ isCore flag).
- **Attendance records**: sessionId + studentId + status (`present | late | absent`) + remarks.
- **Requests**: typed payloads (`course-drop` , `student-registration` , `teacher-registration` , `attendance-update` , `session-edit`) with `approvals` map and `status` .
- **Notifications**: title, description, recipientRole, link.
- **Feedback**: courseId, studentId, rating 1–5, optional comment; used to compute averages.

8) Runbook for common demos

- Reset data: clear `localStorage` keys starting with `nhu-` , refresh.
- Create a new student request: on login page “Request Access”, fill form, submit. Log in as admin → Requests → approve/decline.
- Drop optional course: student Courses → Request Drop → admin approves.
- Attendance change for past class: teacher Take Attendance → pick past session under “Past Classes” → edit → submit → admin approves.
- Session edit for past date: teacher Schedule → Edit → submit → admin approves.
- Ratings: student opens course card → Rate/Feedback; teacher views in Classes; admin edits in Courses table dialog.

9) Troubleshooting

- Node mismatch: ensure `nvm use 20.19.0` (or Volta/asdf pinned). Vite 7 needs Node 20.19+.
- Port busy: Vite auto-increments; read the console URL. Or run `npm run dev -- --port 5175` .
- Blank data / broken nav: clear `localStorage` (Application tab) then refresh.
- Unable to mark attendance: only during live window. For past classes, use the approval path.
- Overlap error adding session: adjust time or date; overlap guard is strict by design.

For quick reference: tech stack is React 19 + TypeScript + Vite 7 + MUI 7 + Redux Toolkit + Day.js; state persists to `localStorage` ; everything is mock/offline-friendly.